Algorithm

Lecture #12

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Heap

Heap

- The number of nodes in a complete binary tree of height h is
- $n = 2^0 + 2^1 + 2^2 + \cdots + 2^h$

$$= \sum_{i=0}^{h} 2^i = 2^{h+1} - 1$$

- h in terms of n is

$h = (\log(n + 1)) - 1 \approx \log n \in \Theta(\log n)$

Heap

Heap

- One of the clever aspects of heaps is that they can be stored in arrays without using any pointers.
- This is due to the left-complete nature of the binary tree.
- We store the tree nodes in level-order traversal.

Heap

Heap

Access to nodes involves simple arithmetic operations:

- left(i): returns 2i, index of left child of node i.
- right(i): returns 2i + 1, the right child.
- parent(i) returns $\lfloor i/2 \rfloor$, the parent of i.

Heap sort Algorithm

Heapsort Algorithm

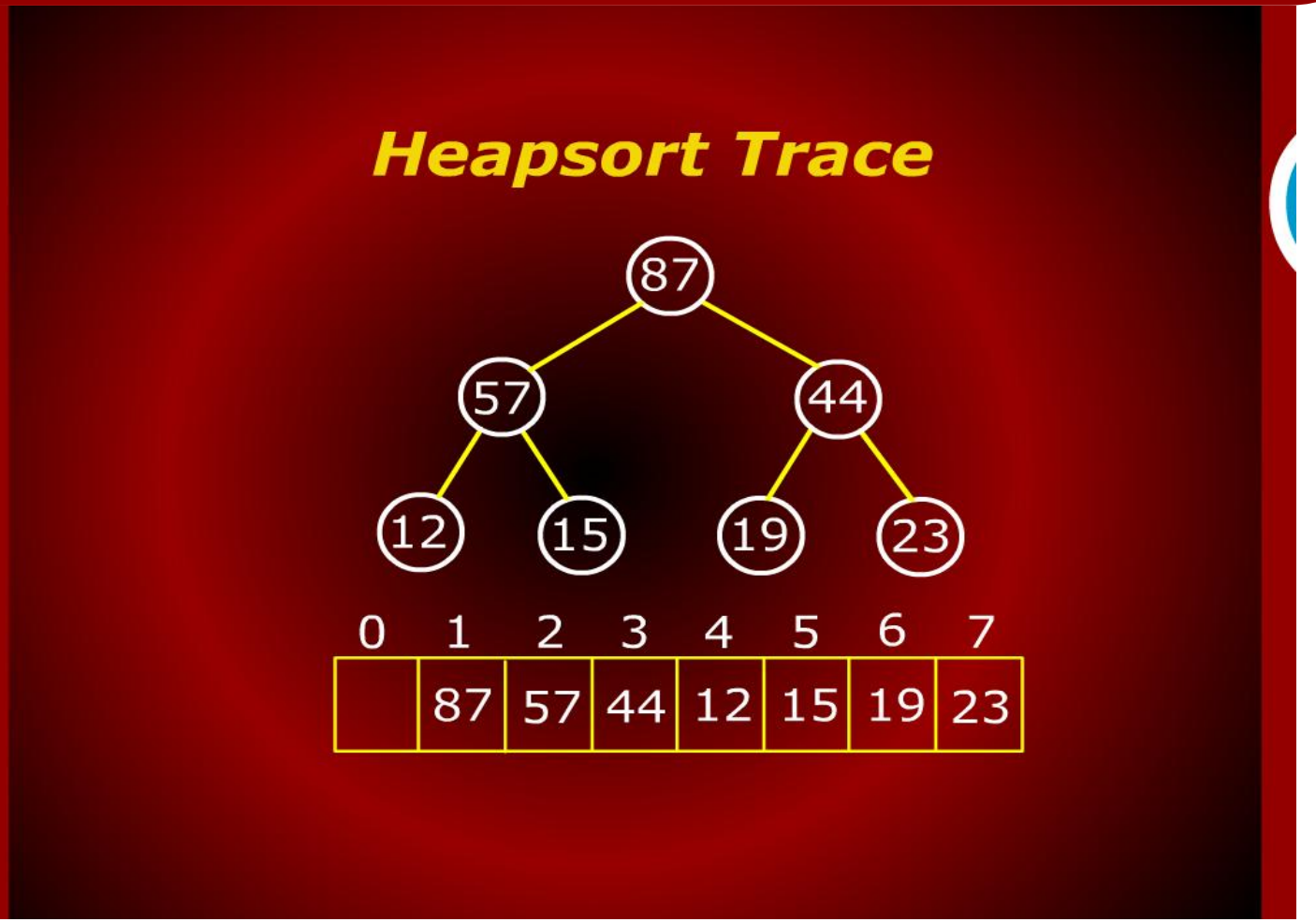
- We build a max heap out of the given array of numbers A[1..n].
- We repeatedly extract the the maximum item from the heap.
- Once the max item is removed, we are left with a hole at the root.
- To fix this, we will replace it with the last leaf in tree.

Heap sort Trace

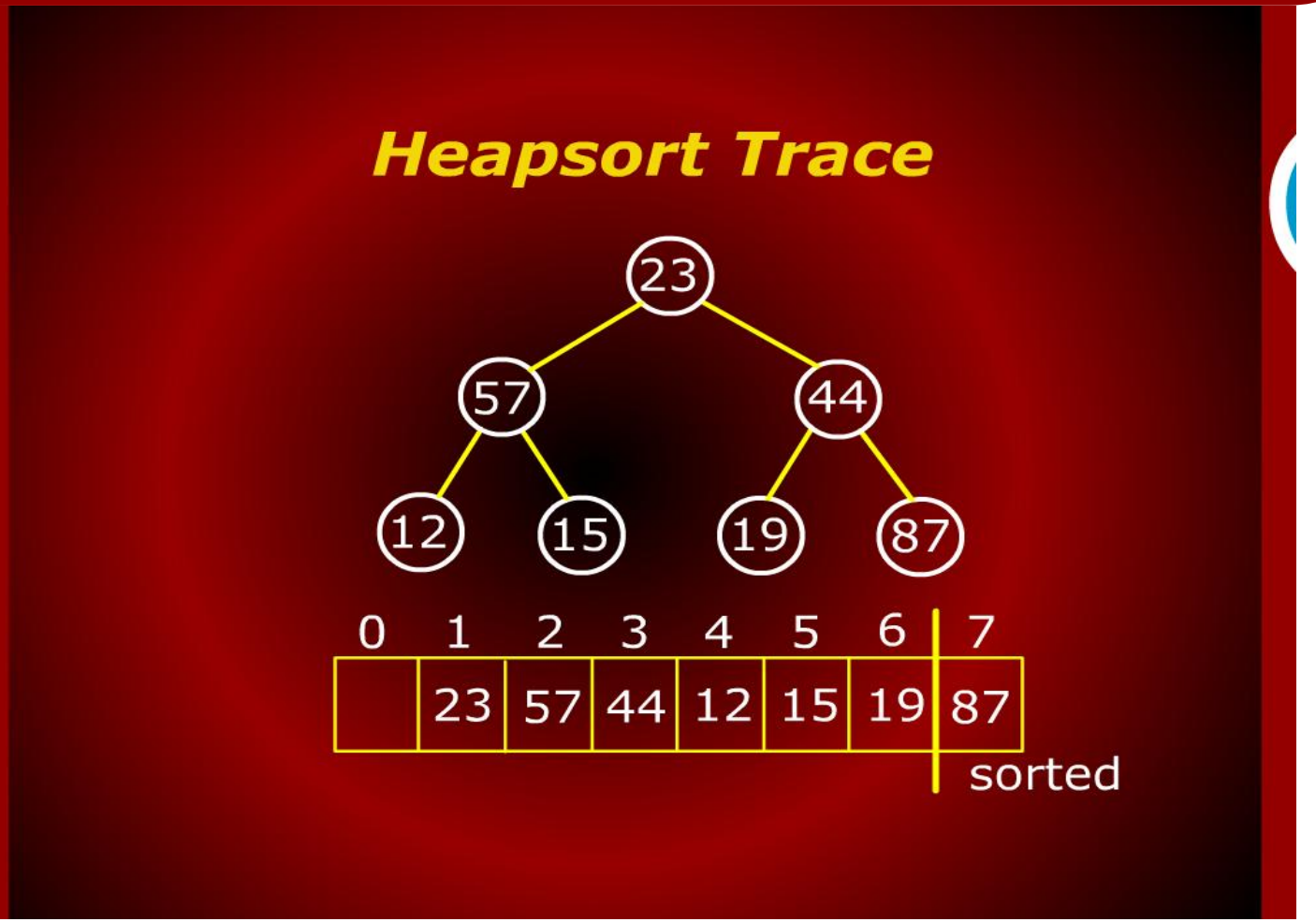
Heap Sort

```
HEAPSORT( array A, int n)
 1 BUILD-HEAP(A, n)
 2 m ← n
 3 while (m ≥ 2)
 4 do SWAP(A[1],A[m])
 5      m ← m – 1
 6      HEAPIFY(A, 1,m)
```

Heap sort Trace



Heap sort Trace



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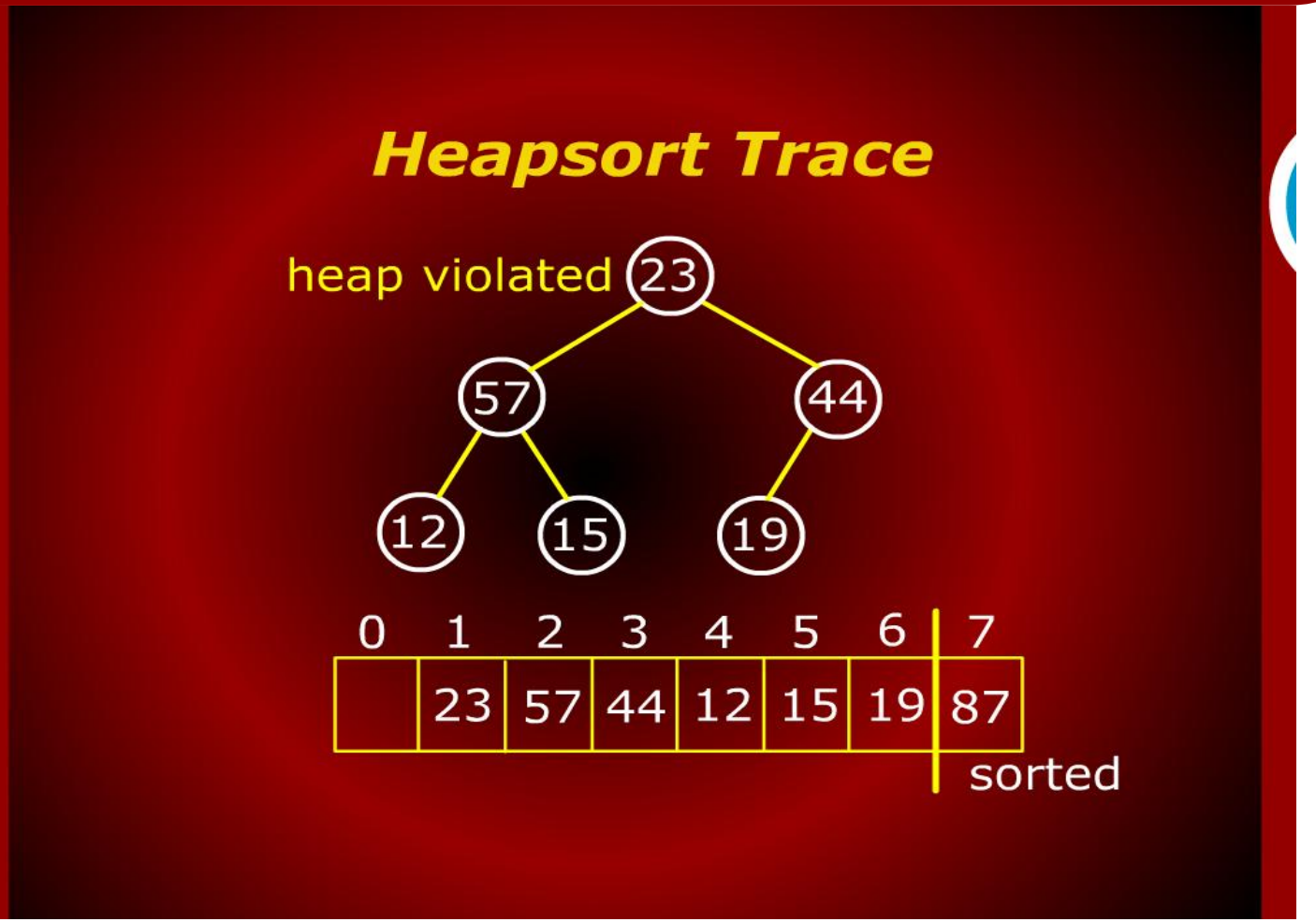

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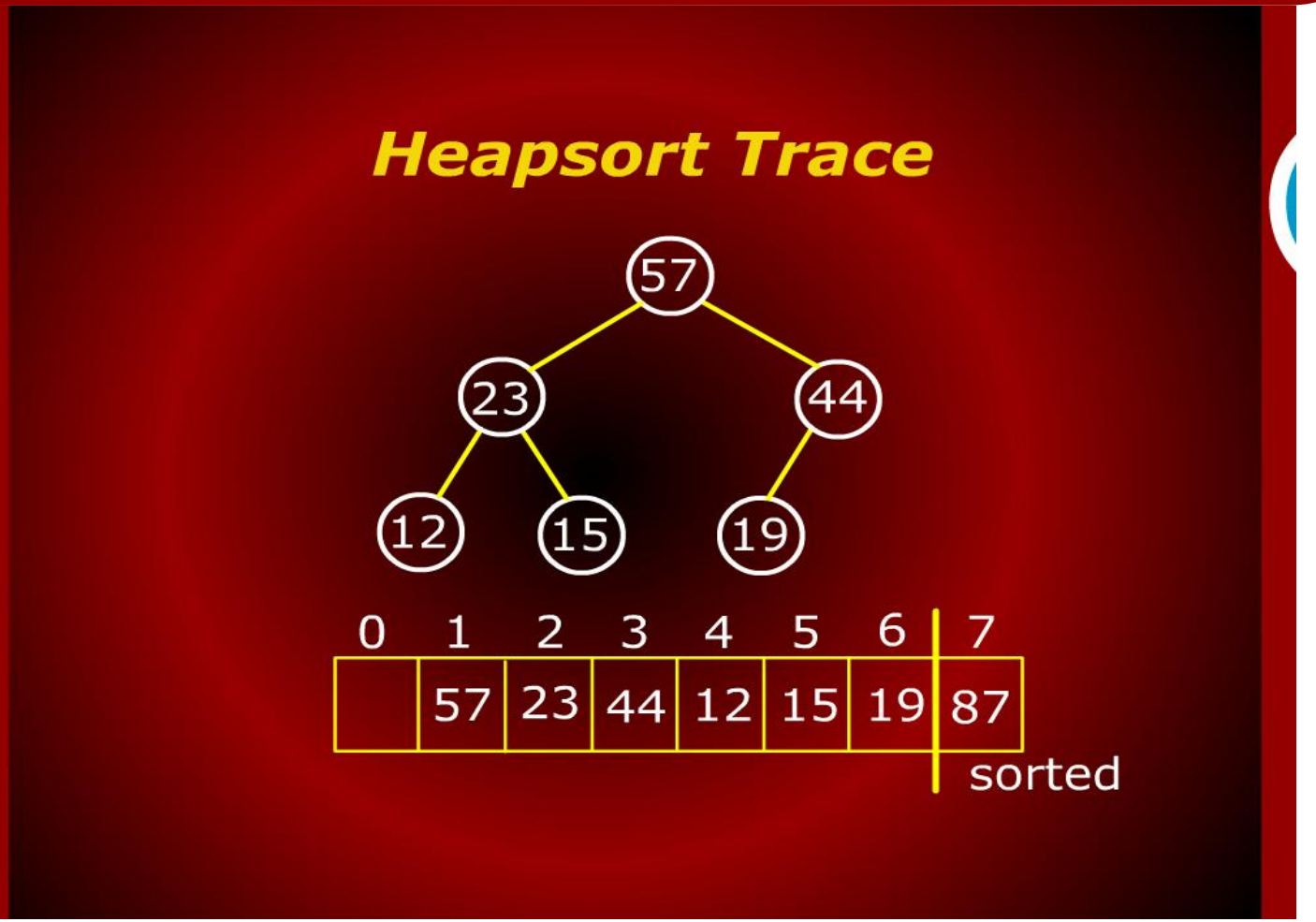

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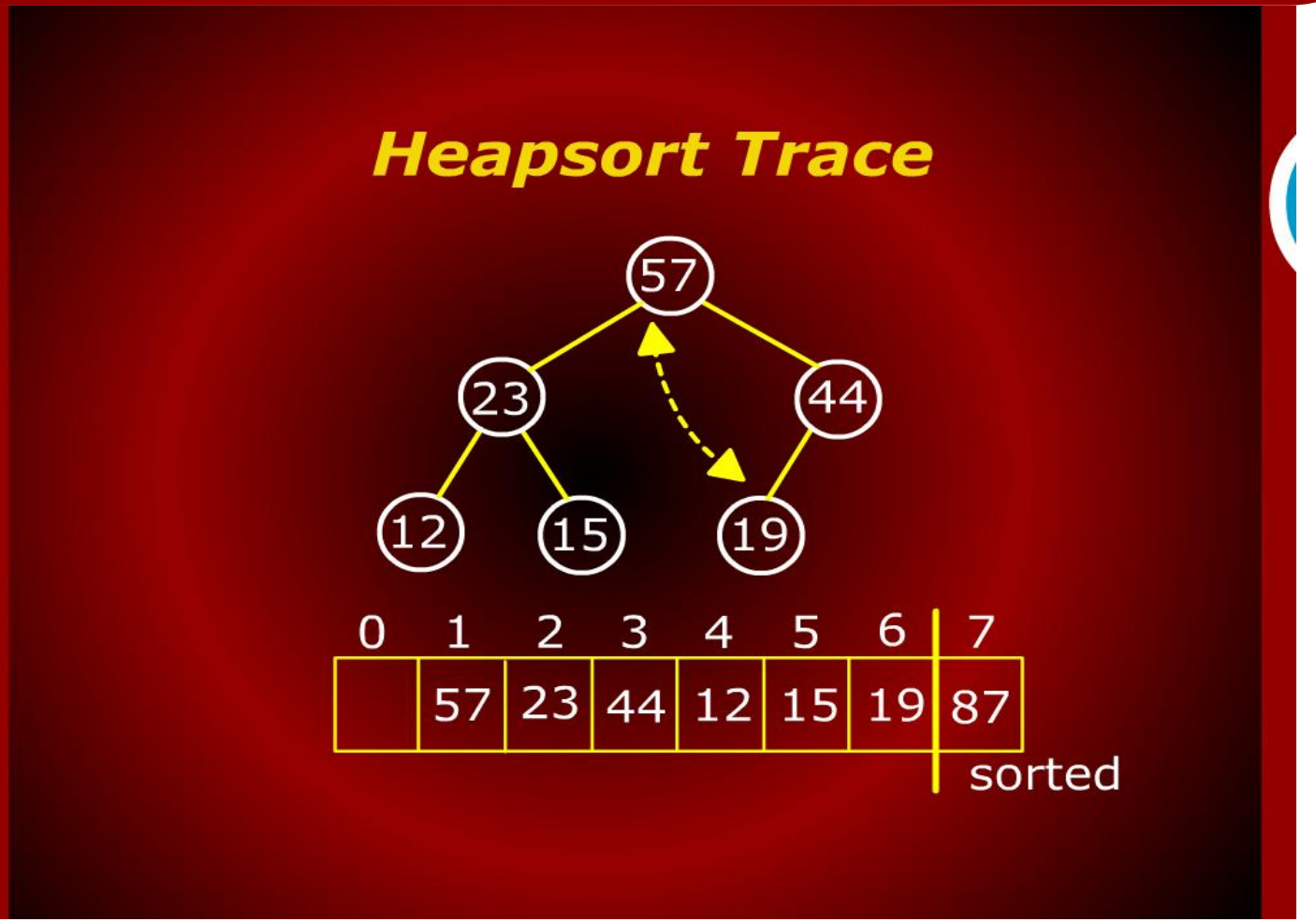

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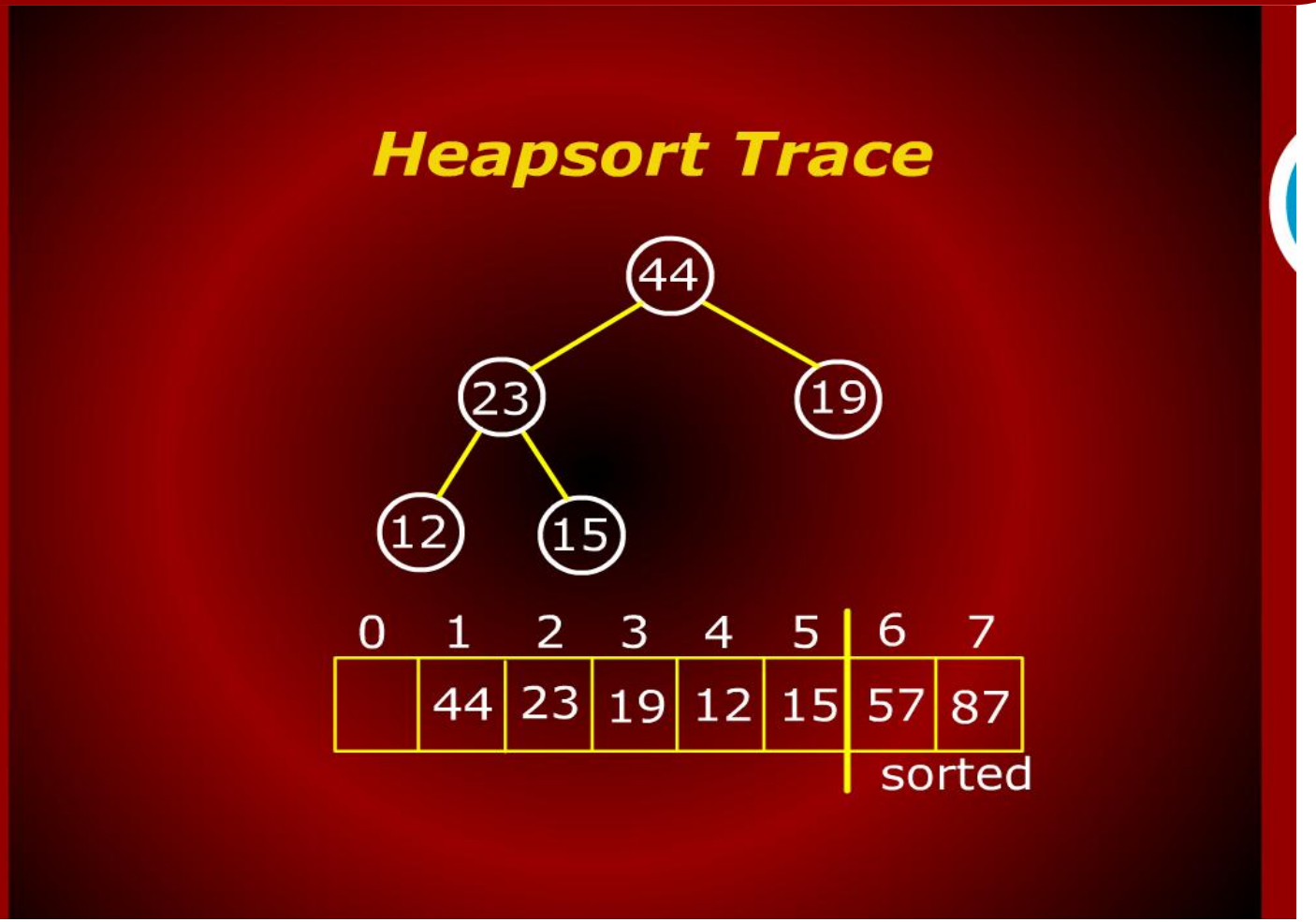

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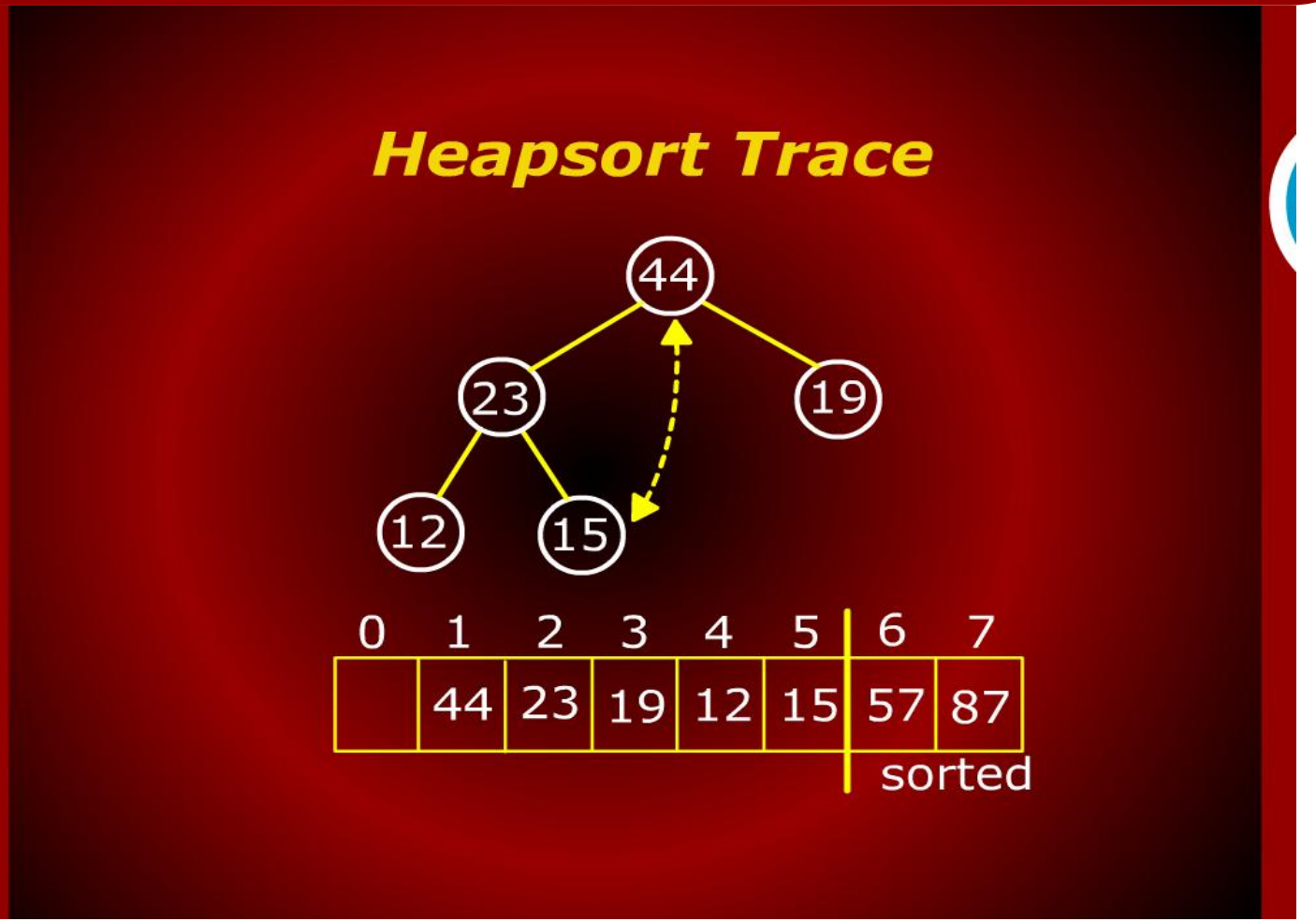

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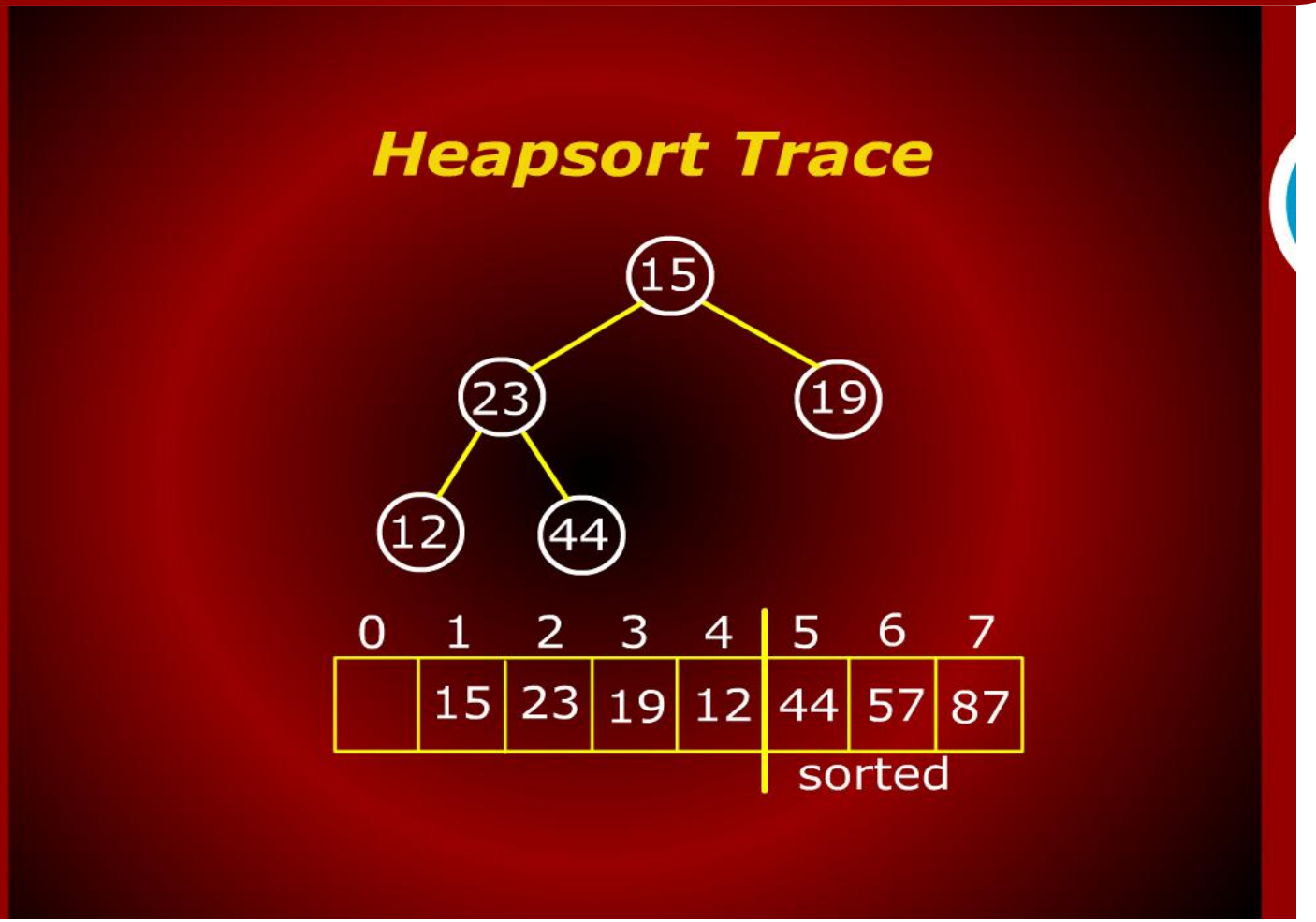

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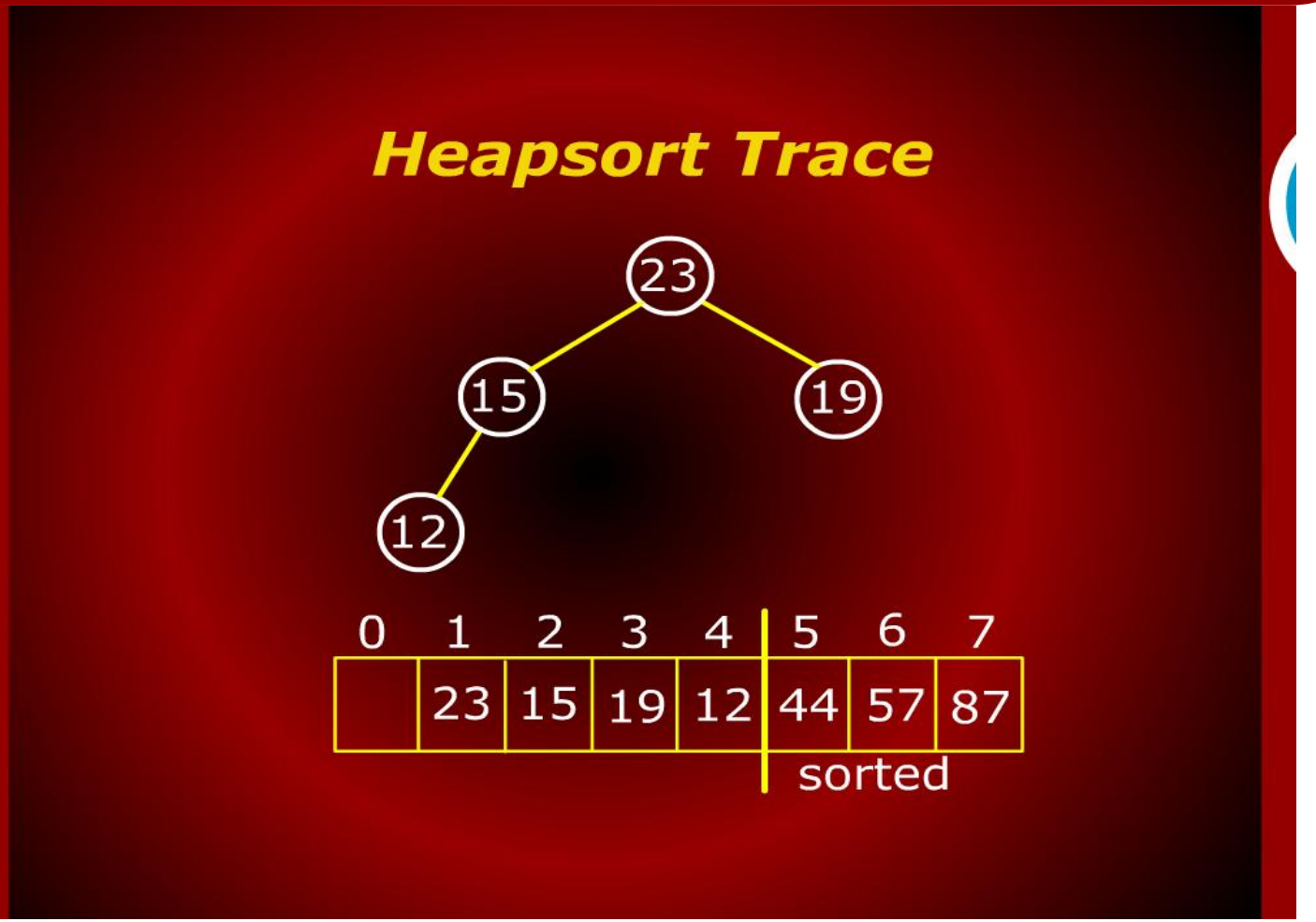

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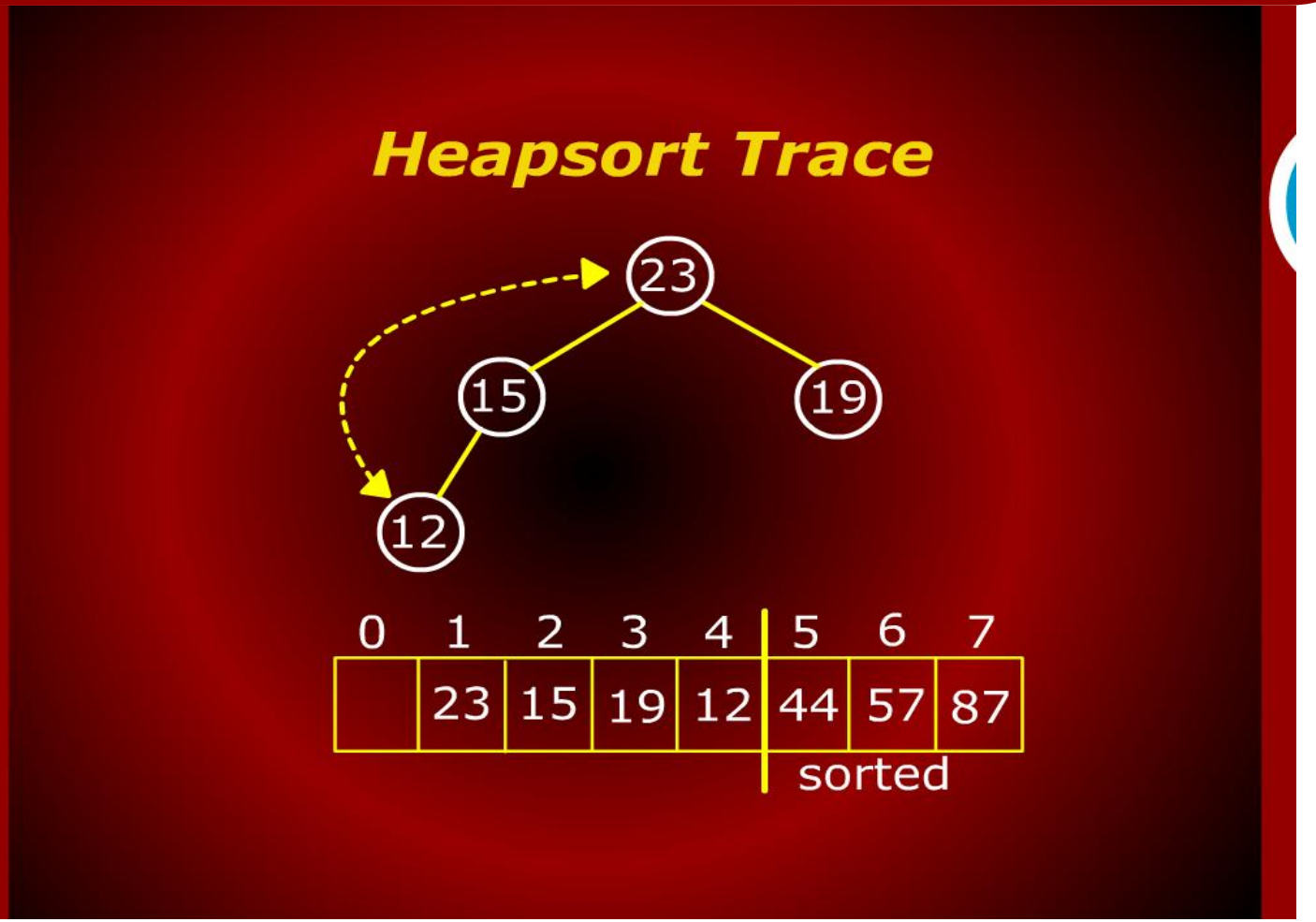

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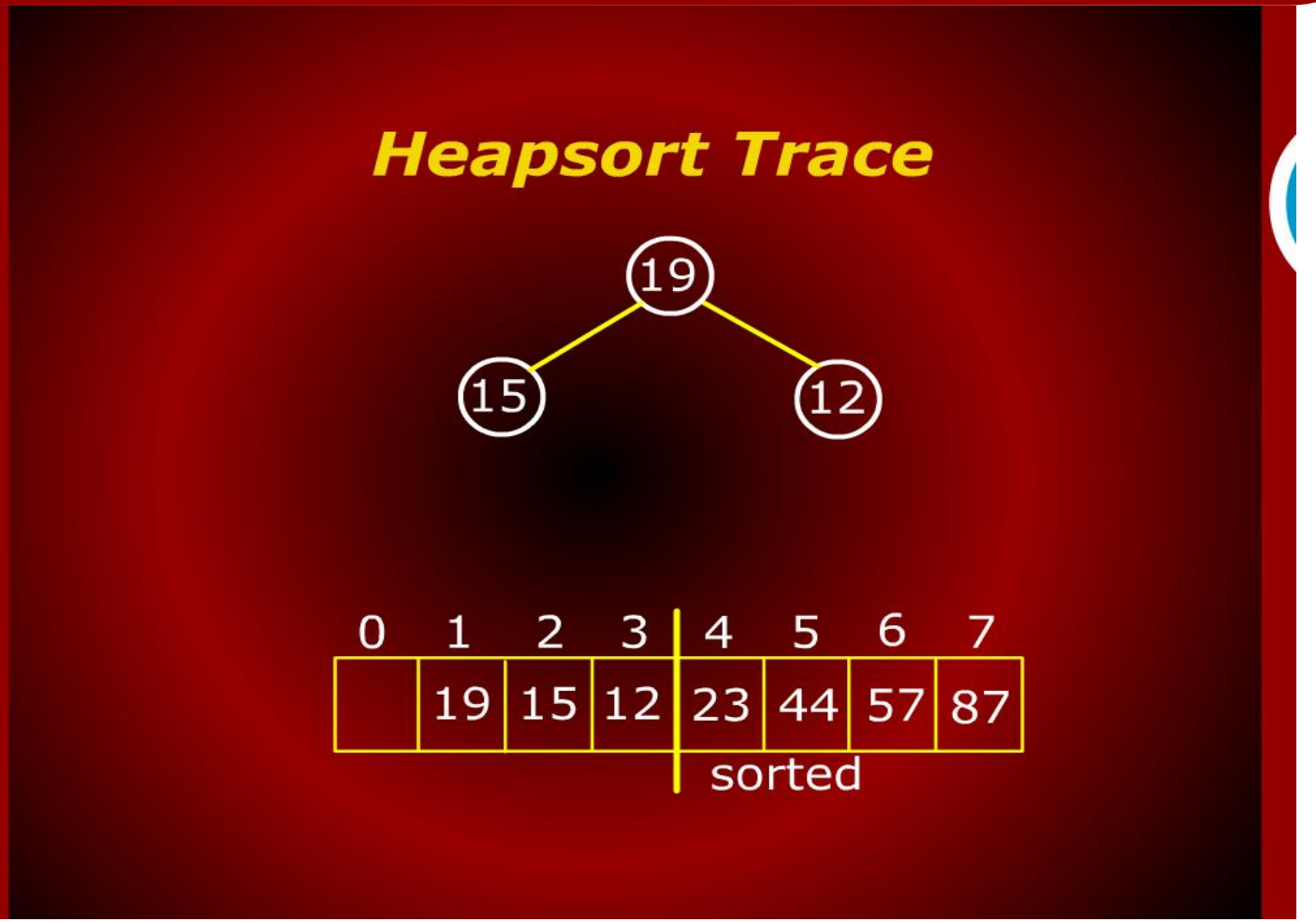

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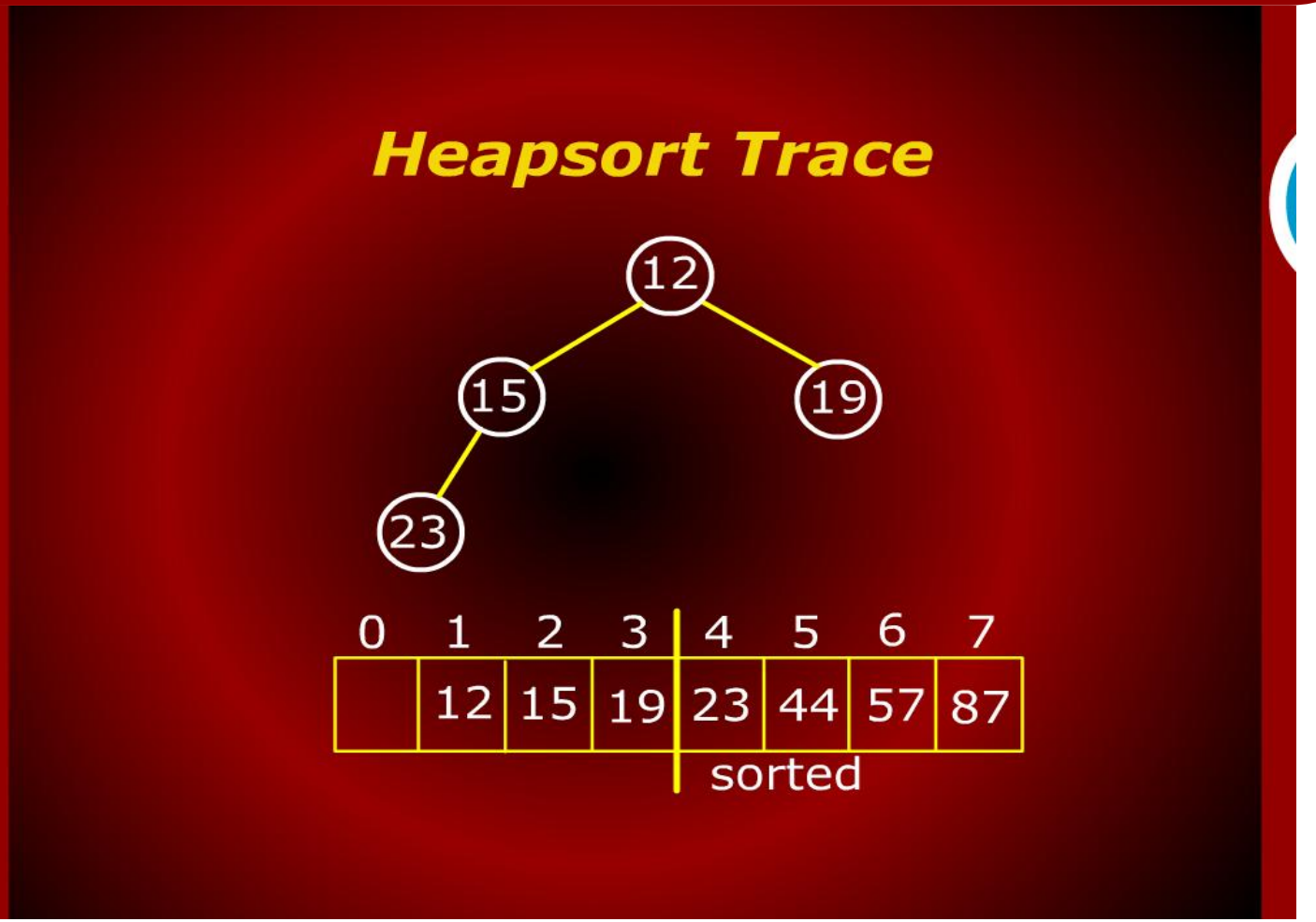

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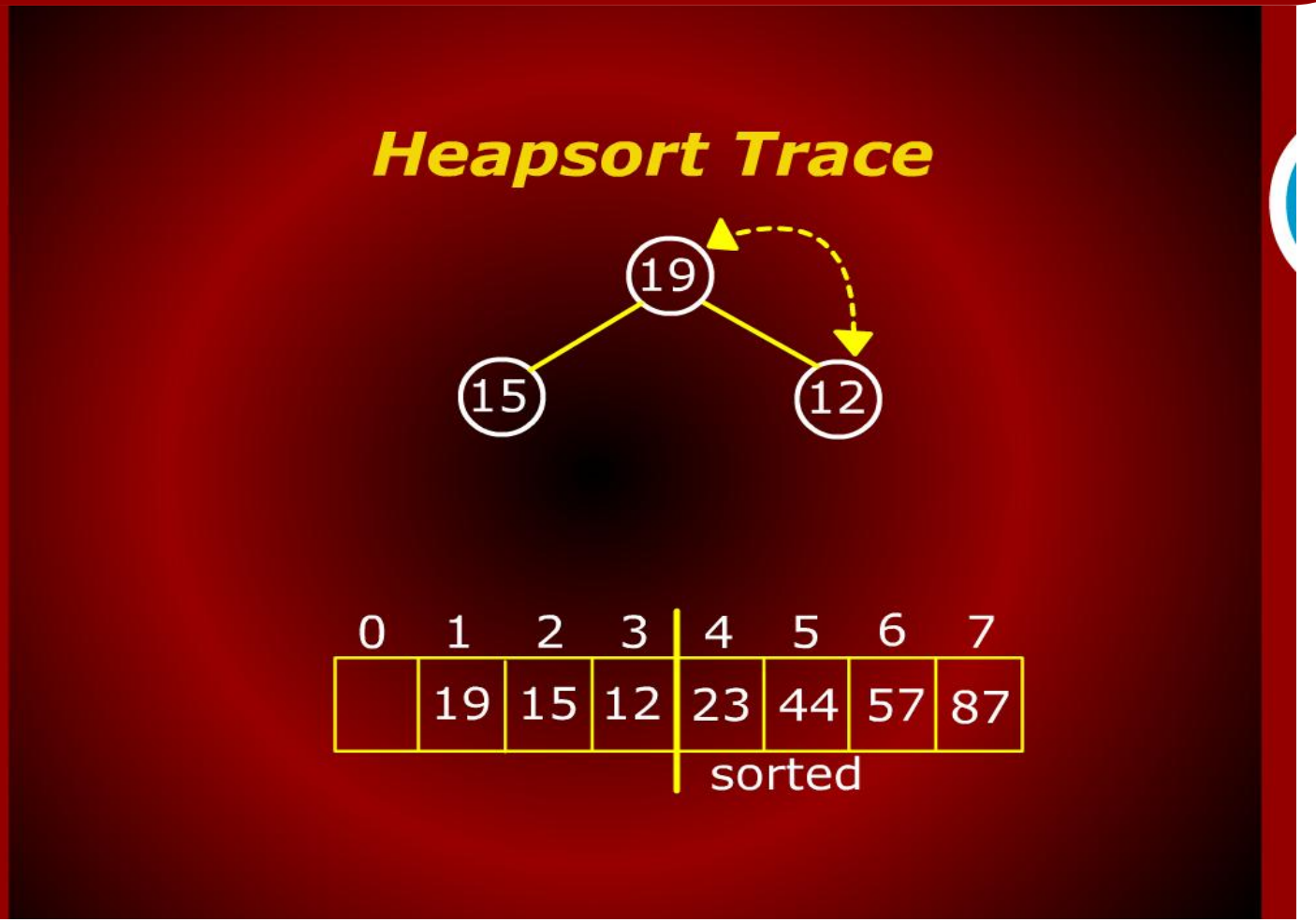

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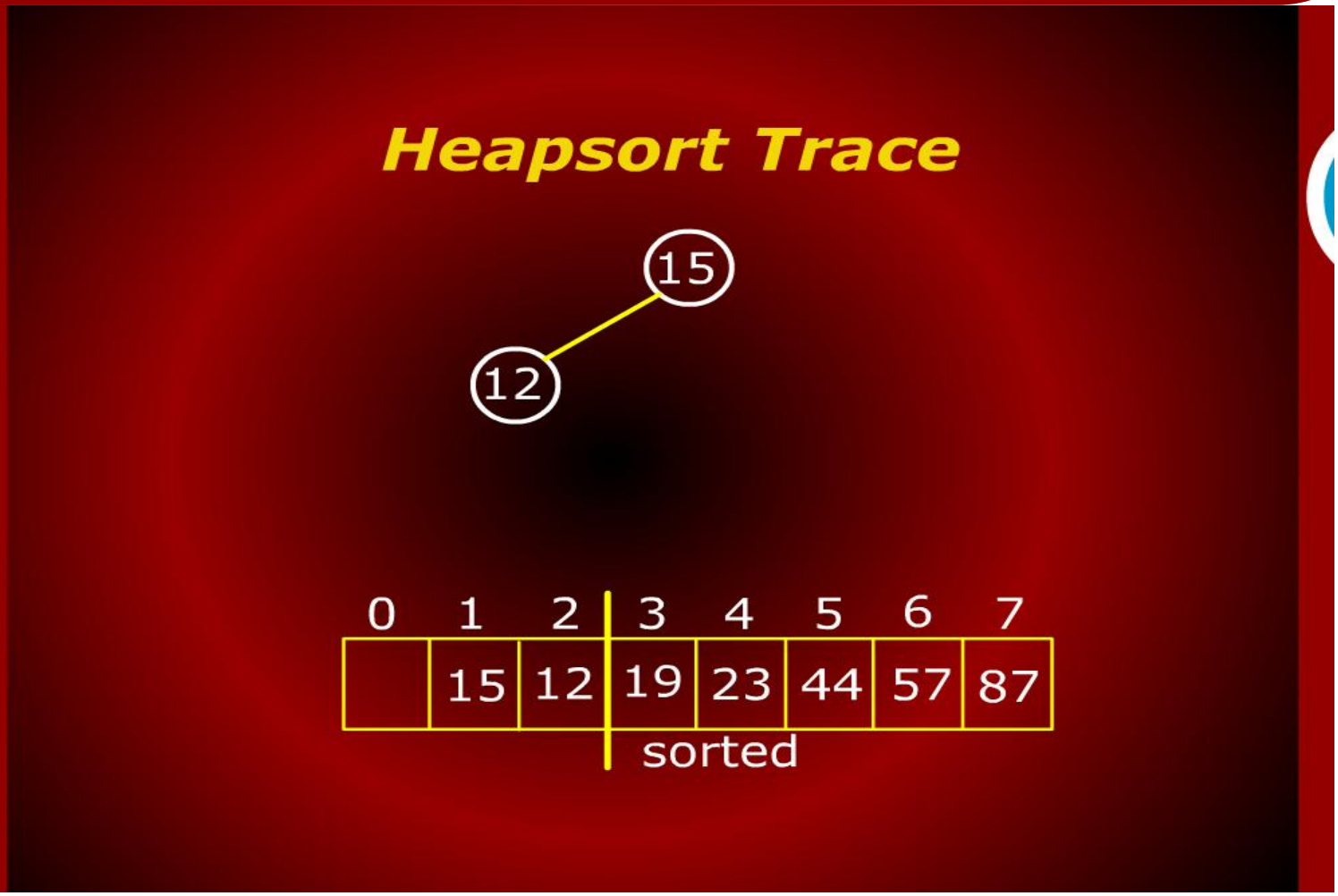

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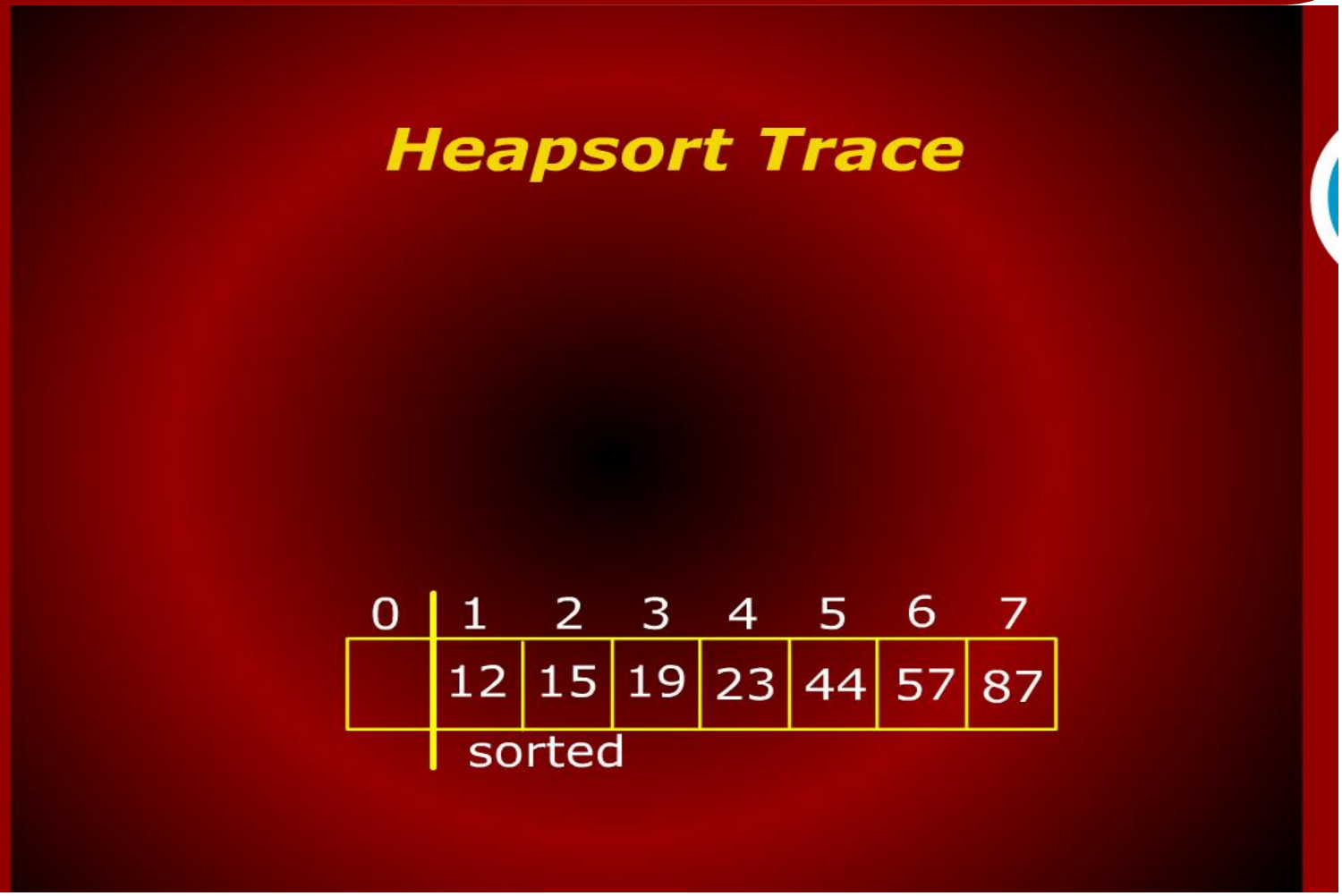
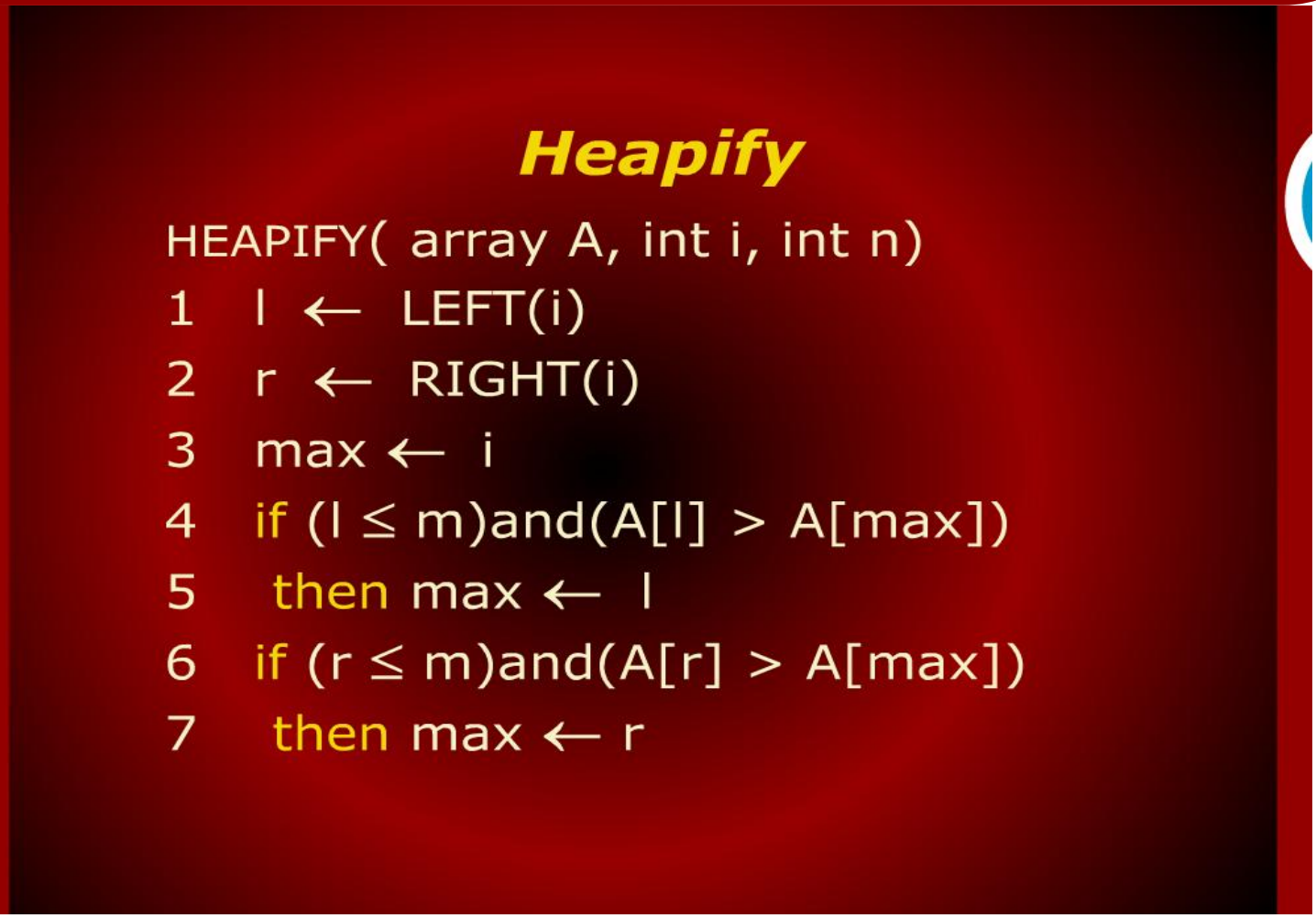


Heapify


Heapify
HEAPIFY(array A, int i, int n)
1 l ← LEFT(i)
2 r ← RIGHT(i)
3 max ← i
4 if (l ≤ m)and(A[l] > A[max])
5 then max ← l
6 if (r ≤ m)and(A[r] > A[max])
7 then max ← r

Heapify

Heapify

```
3   max ← i
4   if (l ≤ m)and(A[l] > A[max])
5    then max ← l
6   if (r ≤ m)and(A[r] > A[max])
7    then max ← r
8   if (max ≠ i)
9   then SWAP(A[i],A[max])
10 HEAPIFY(A,max,m)
```

Analysis of Heapify

Analysis of Heapify

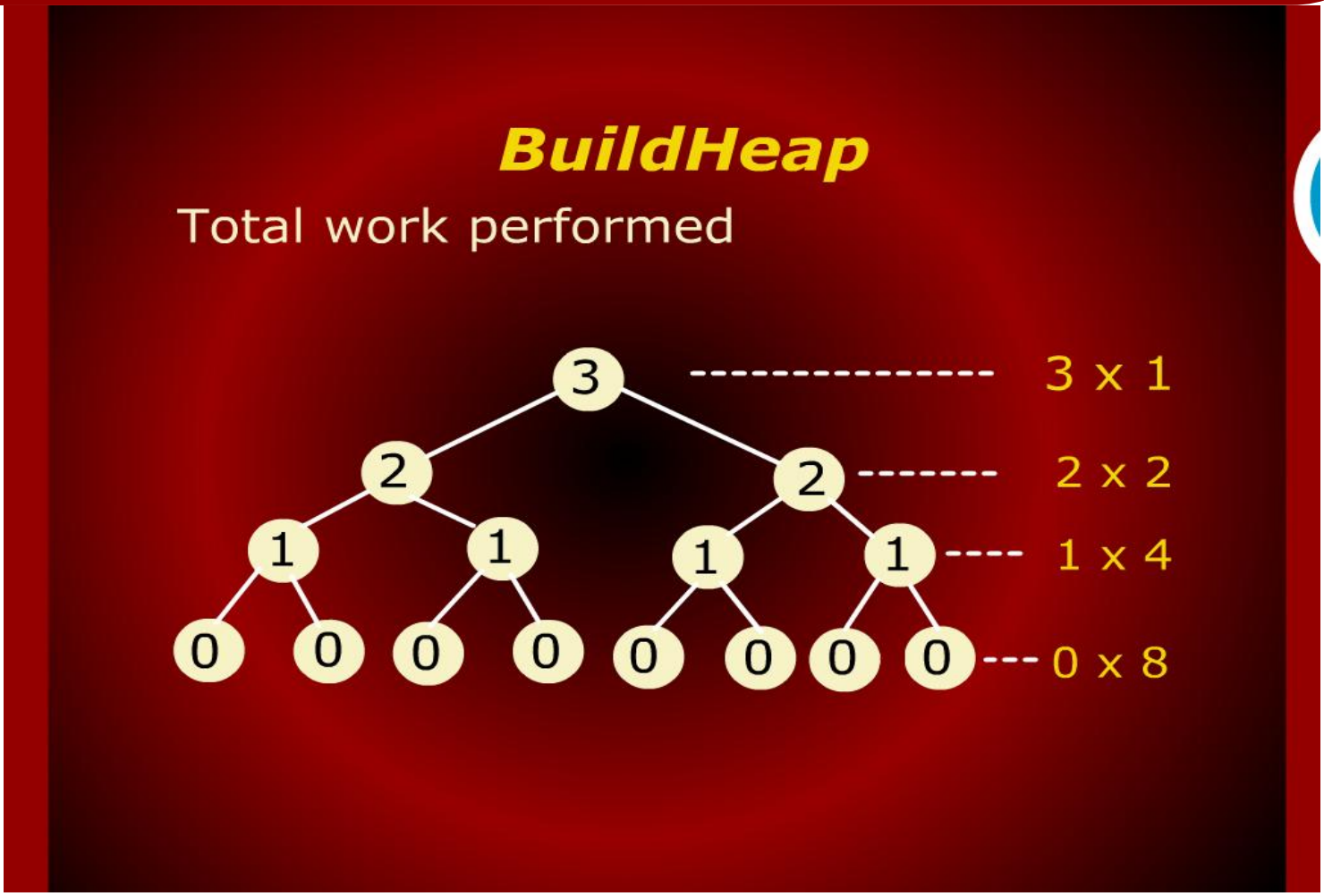
- We call heapify on the root of the tree.
- The maximum levels an element could move up is $\Theta(\log n)$ levels.
- At each level, we do simple comparison which $O(1)$.
- The total time for heapify is thus $O(\log n)$.
- Notice that it is not $\Theta(\log n)$ since, for example, if we call heapify on a leaf, it will terminate in $\Theta(1)$ time.

Analysis of BuildHeap

Analysis of BuildHeap

- For convenience, we will assume $n = 2^{h+1} - 1$ where h is the height of tree..
- The heap is a left-complete binary tree.
- Thus at each level j, $j < h$, there are 2^j nodes in the tree.
- At level h, there will be 2^h or less nodes.
- How much work does buildHeap carry out?

BuildHeap

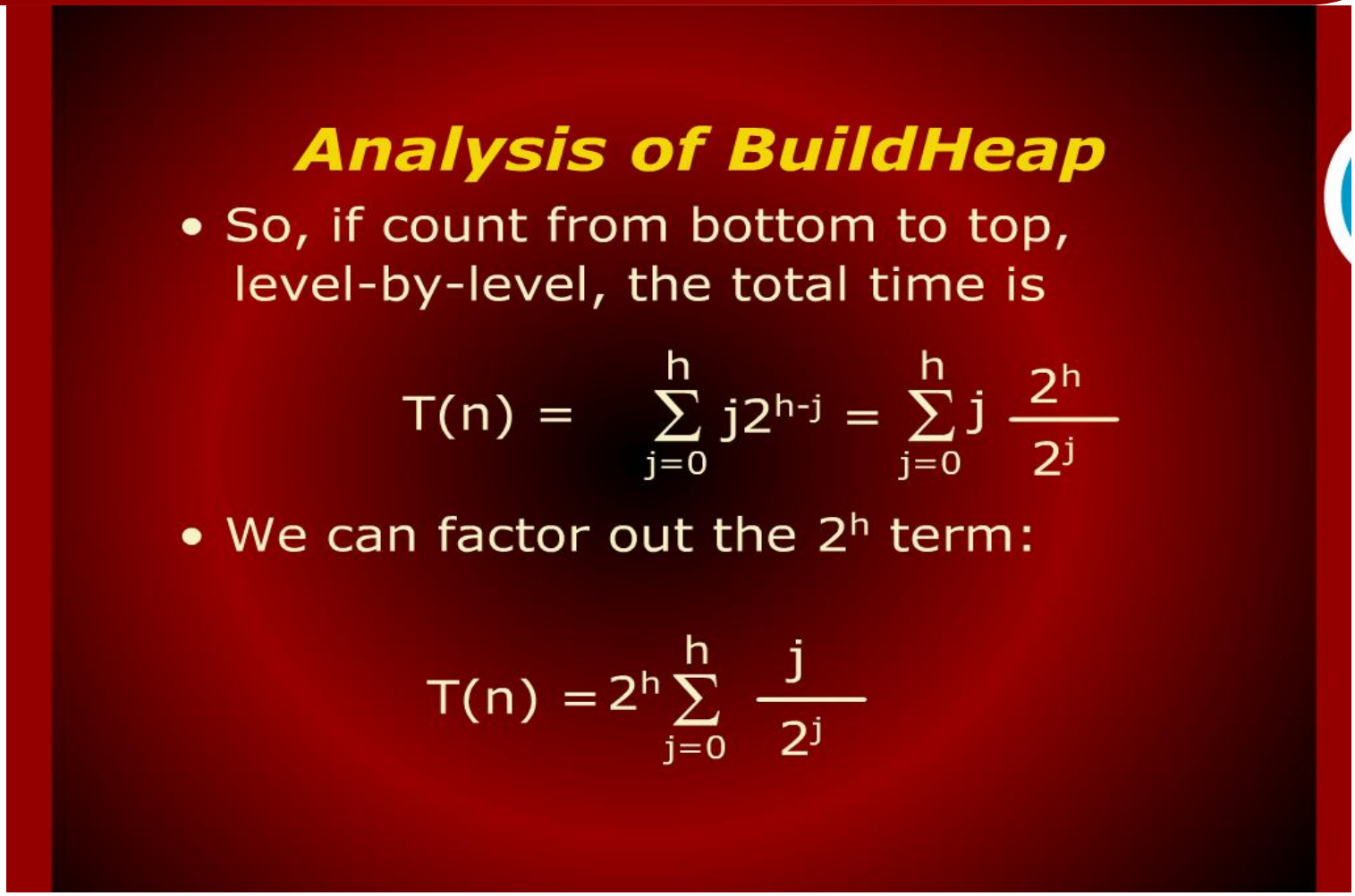

BuildHeap
Total work performed
3
2
2
1
1
1
1
0
0
0
0
0
0
0
0
3 x 1
2 x 2
1 x 4
0 x 8

Analysis of BuildHeap

Analysis of BuildHeap

- At the bottom most level, there are 2^h nodes but we do not heapify these.
- At the next level up, there are 2^{h-1} nodes and each might shift down 1.
- In general, at level j, there are 2^{h-j} nodes and each may shift down j levels.

Analysis of BuildHeap



Analysis of BuildHeap

Analysis of BuildHeap

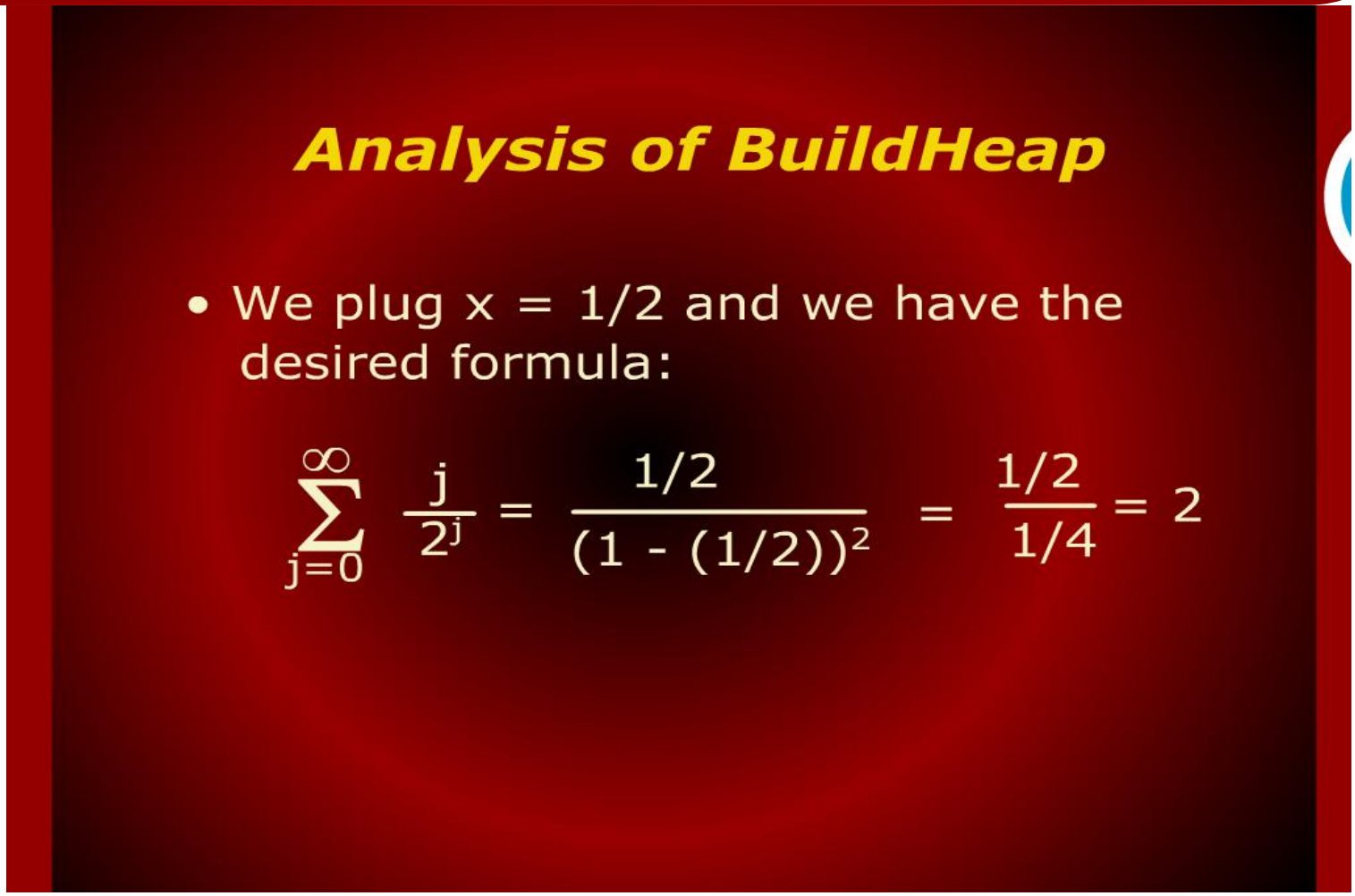
- How do we solve this sum? Recall the geometric series, for any constant $x < 1$

$$\sum_{j=0}^{\infty} x^j = \frac{1}{1-x}$$

- Take the derivative with respect to x and multiply by x

$$\sum_{j=0}^{\infty} jx^{j-1} = \frac{1}{(1-x)^2} \qquad \sum_{j=0}^{\infty} jx^j = \frac{x}{(1-x)^2}$$

Analysis of BuildHeap



Analysis of BuildHeap

Analysis of BuildHeap

- In our case, we have a bounded sum, but since the infinite series is bounded, we can use it as an easy approximation:

$$T(n) = 2^h \sum_{j=0}^{h} \frac{j}{2^j}$$

$$\leq 2^h \sum_{j=0}^{\infty} \frac{j}{2^j}$$

$$\leq 2^h \cdot 2 = 2^{h+1}$$

Analysis of BuildHeap

Analysis of BuildHeap

- Recall that $n = 2^{h+1} - 1$.
- Therefore

$$T(n) \leq n + 1 \in O(n)$$

- The algorithm takes at least $\Omega(n)$ time since it must access every element at once.
- So the total time for BuildHeap is $\Theta(n)$.

Analysis of BuildHeap

Analysis of BuildHeap

- BuildHeap is a relatively complex algorithm.
- Yet, the analysis yield that it takes $\Theta(n)$ time.
- An intuitive way to describe why it is so is to observe an important fact about binary trees
- The fact is that the vast majority of the nodes are at the lowest level of the tree.

Analysis of BuildHeap

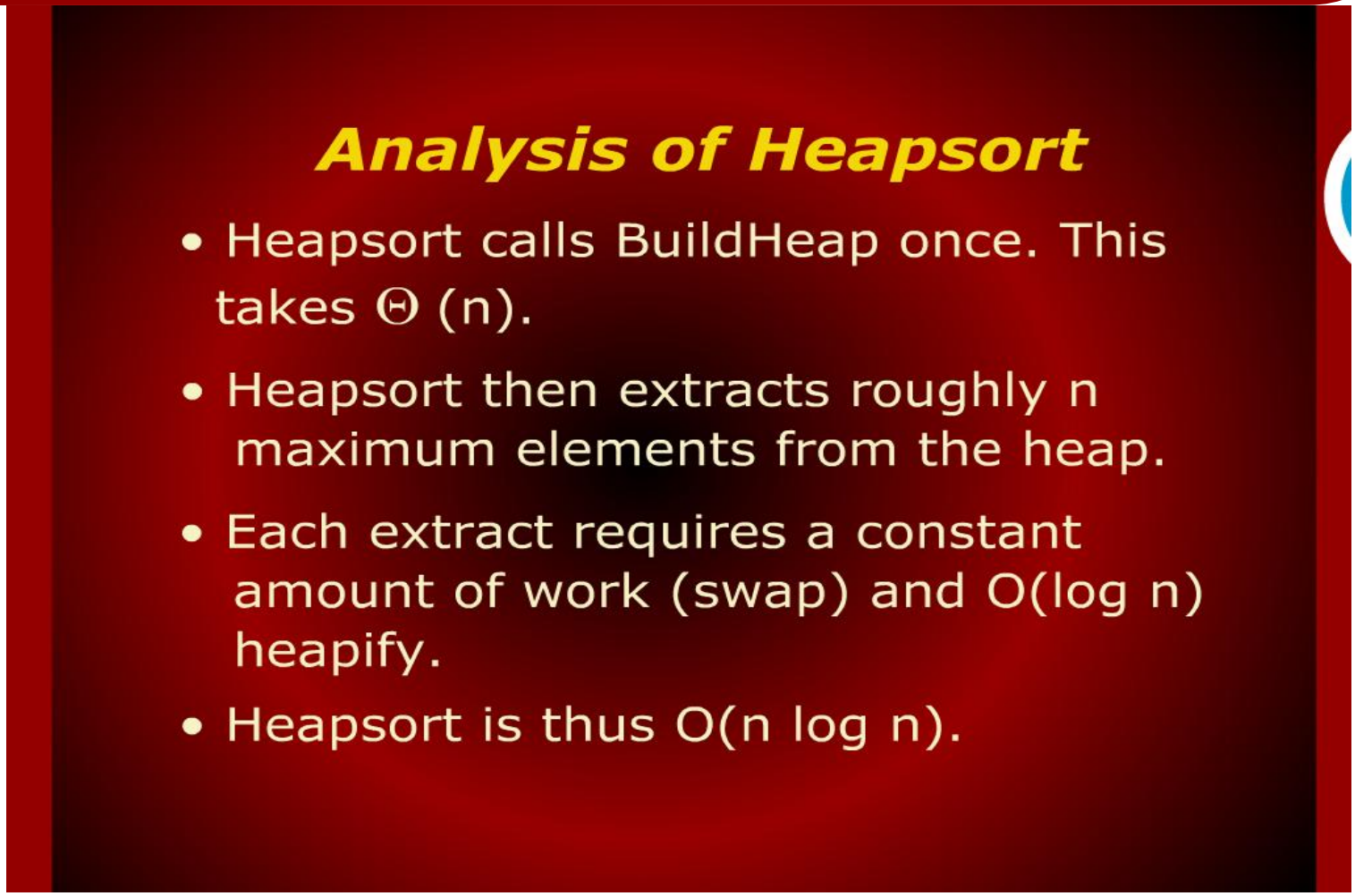
Analysis of BuildHeap

- The number of nodes at the bottom three levels alone is

$$2^h + 2^{h-1} + 2^{h-2} = \frac{n}{2} + \frac{n}{4} + \frac{n}{8} = \frac{7n}{8} = 0.875n$$

- Almost 90% of the nodes of a complete binary tree reside in the 3 lowest levels.
- Thus, algorithms that operate on trees should be efficient (as BuildHeap is) on the bottom-most levels since that is where most of the weight of the tree resides.

Analysis of Heapsort



Analysis of Heapsort

Analysis of Heapsort

- Is HeapSort $\Theta(n \log n)$?
- The answer is yes.
- In fact, later we will show that comparison based sorting algorithms can not run faster than $\Omega(n \log n)$.
- Heapsort is such an algorithm and so is Mergesort that we saw ealier.